

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12 ASSESSMENT TASK

EXTENSION 2 MATHEMATICS

MARCH 2005

Instructions

- * Attempt all questions.
- * Answers to be written on the paper provided.
- * Start each question on a new page.
- * Marks may not be awarded for careless or badly arranged working.
- * Indicated marks are a guide and may be changed slightly if necessary.
- * These questions must be handed in attached to the top of your solutions.

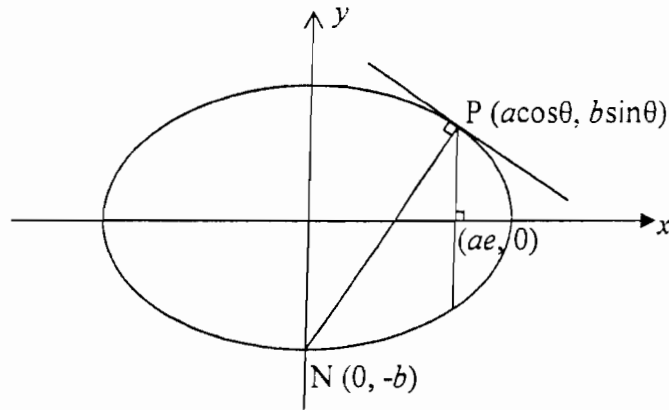
Q1 /16	Q2 /17	Q3 /18	TOTAL
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QUESTION 1

- a) Find $|(3 - 4i)^n|$ (2)
- b) (i) On an Argand diagram shade in the region determined by the inequalities
 $2 \leq \text{Im}(z) \leq 4$ and $\frac{\pi}{6} \leq \arg(z) \leq \frac{\pi}{4}$. (3)
- (ii) Let z_0 be the complex number of maximum modulus satisfying the inequalities in (i). Express z_0 in the form $x + iy$. (1)
- c) Find pairs of integers x and y which satisfy the condition
 $(x + iy)^2 = -3 - 4i$. (3)
- d) If $z = \cos \theta + i \sin \theta$ use De Moivre's Theorem or otherwise to simplify
 $z^4 + \frac{1}{z^4}$. (2)

Question 1 (Cont)

e)



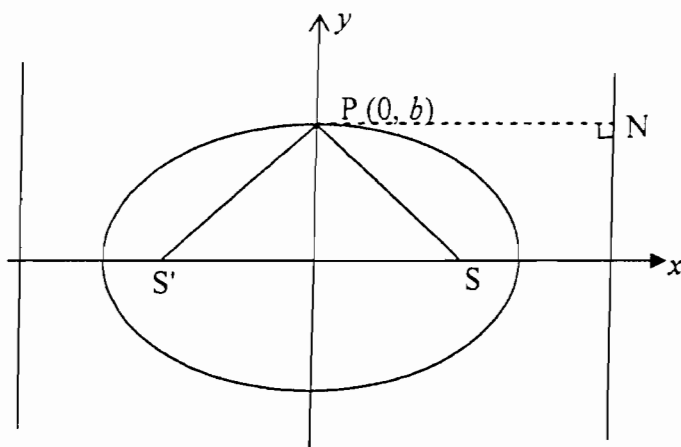
The chord through the focus $(ae, 0)$ of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at right angles to the x -axis meets the ellipse at $P (a \cos \theta, b \sin \theta)$. The normal at P passes through the point $(0, -b)$.

(i) Show that $\cos \theta = e$ and $\sin \theta = \sqrt{1 - e^2}$. (2)

(ii) Given the equation of the normal at P is $ax \sin \theta - by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$, show that the condition for it to pass through $(0, -b)$ is $e^4 + e^2 - 1 = 0$.
(You may show instead that $e^6 - 2e^2 + 1 = 0$, which is another version of the above condition) (3)

QUESTION 2

a)



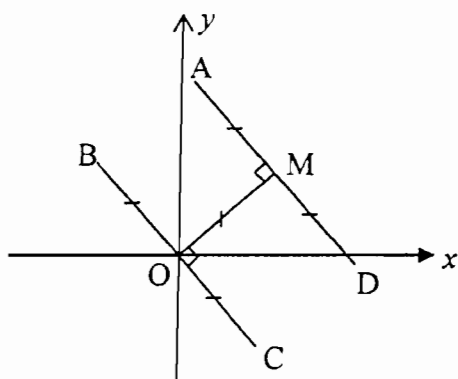
If $P(0, b)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where S and S' are the foci and N is a point on the directrix.

(i) Write down the value of the ratio $\frac{PS}{PN}$. (1)

(ii) Hence or otherwise show that $PS + PS' = 2a$. (2)

(iii) Explain why the perimeter of the triangle PSS' is always less than $4a$ units. (2)

b)



In the diagram $AM = MD = OM = OB = OC$ and $AD \perp OM \perp BC$. O is the origin.

If M represents the complex number z

(i) Which point represents the complex number iz ? (1)

(ii) Find, in terms of z , the complex number represented by the point D . (2)

Question 2 (Cont)

- c) (i) Sketch the curve $y = (x - 1)^2$ and shade the region bounded by the curve, the x axis and the line $x = 2$. (1)
- (ii) The region in (i) is rotated about the line $y = -1$. Find the volume of the solid formed by this rotation. (3)
- d) (i) Sketch the locus of the complex number z if $|z - 1| = 1$. (1)
- (ii) Let z be a complex number which satisfies the locus in (i) and let $\arg(z) = \theta$. Explain with the aid of your graph or otherwise why $\arg(z - 1) = 2\theta$. (2)
- (iii) Find $\arg(z^2 - 3z + 2)$ in terms of θ . (2)

QUESTION 3

- a) (i) Express $z = \sqrt{3} + i$ in modulus/argument form. (2)
- (ii) Show that z is a complex solution of the equation $x^7 + 64x = 0$. (2)
- b) If $z = x + iy$
- (i) Write $\frac{1}{z}$ as a complex number. (1)
- (ii) Hence find the equations of the locus of z if $\operatorname{Re}\left(z - \frac{1}{z}\right) = 0$. (2)
- c) If the roots of the equation $z^8 = 1$ are $1, w, w^2, w^3, w^4, w^5, w^6, w^7$ where w is the complex root with the smallest positive argument
- (i) Find w^3 in mod-arg form. (1)
- (ii) Evaluate $w^2 + w^4 + w^6$ giving a reason. (2)

Question 3 (Cont).

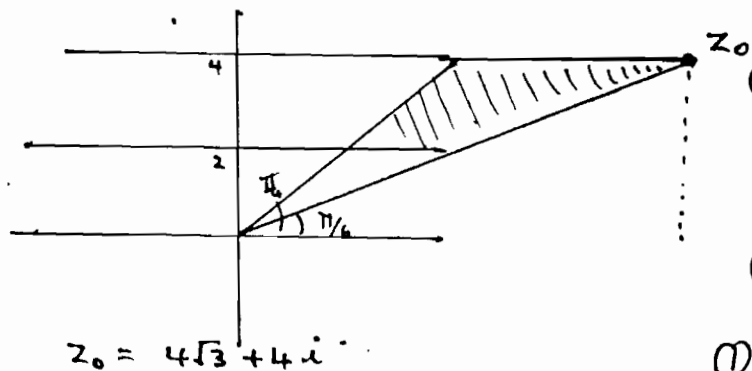
- d) (i) Differentiate $\frac{x^2}{25} + \frac{y^2}{9} = 1$ implicitly. (2)
- (ii) Derive the equation of the tangent to the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ at (x_1, y_1) . (2)
- (iii) Write down the equations of the directrices. (1)
- (iv) If $x_1 > 0$ and $y_1 > 0$ find the values of x_1 so that the tangent at (x_1, y_1) intersects the nearest directrix below the x axis. (3)

End of Exam

①

a) $|3 - 4i| = \sqrt{3^2 + 4^2}$
 $= 5$
 $|3 - 4i|^n = 5^n$ — ①

b)



① - understanding
 Im
 ① - understanding
 arg
 ① - region.

$z_0 = 4\sqrt{3} + 4i$

①

c) $(x + iy)^2 = x^2 - y^2 + 2ixy$
 Now $x^2 - y^2 = -3$
 $2xy = -4$ — ① equating
 $xy = -2$
 $x = -1, y = 2$ or $x = 1, y = -2$ ① each answer.

d) $z^4 = \cos 4\theta + i \sin 4\theta$ — ①
 $z^{-4} = \cos(4\theta) + i \sin(4\theta)$
 $= \cos 4\theta - i \sin 4\theta$ — ①
 $\therefore z^4 + z^{-4} = 2\cos 4\theta$ — ①

e) i) comparing x-value of P and S
 $ae = a \cos \theta$
 $\therefore \cos \theta = e$ — ①
 Now $\sin^2 \theta + \cos^2 \theta = 1$
 $\therefore \sin^2 \theta = 1 - e^2$
 $\sin \theta = \sqrt{1 - e^2}$ — ①

ii) at $(0, -b)$
 $b^2 \cos \theta = (a^2 - b^2) \cos \theta \sin \theta$
 now $\cos \theta = e$ and $\sin \theta = \sqrt{1 - e^2}$

$\therefore b^2 e^2 = (a^2 - b^2) e \sqrt{1 - e^2}$
 for the ellipse $b^2 = a^2(1 - e^2)$ — ①

$a^2(1 - e^2)e = [a^2 - a^2(1 - e^2)] e \sqrt{1 - e^2}$

$a^2 e(1 - e^2) = a^2 e^3 \sqrt{1 - e^2}$

since $0 < e < 1$ for ellipse

$1 - e^2 = e^2 \sqrt{1 - e^2}$

$\therefore (1 - e^2)^2 = e^4(1 - e^2)$
 $1 - 2e^2 + e^4 = e^4 - e^6$

$\therefore e^6 - 2e^2 + 1 = 0$

② for work

Question 2

a) i) e

ii) Now $PS = e PN$
and $PS' = e PN'$ (mark on diagram) — ①

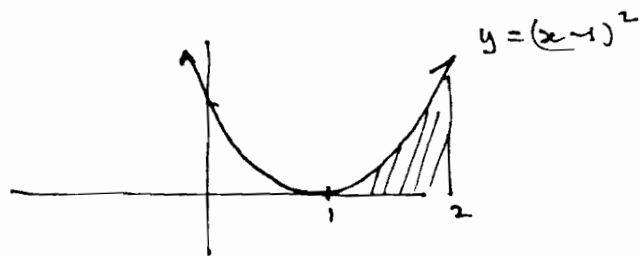
$$\begin{aligned}\therefore PS + PS' &= e[PN + PN'] \\ &= e\left[\frac{a}{e} + \frac{a}{e}\right] \\ &= 2a\end{aligned}$$

iii Perimeter $PS'S = 2a + 2ae$
but $e < 1$ for ellipse
 $\therefore PS'S < 4a$

b) i) B

$$\begin{aligned}\text{ii } \vec{OD} &= \vec{OM} + \vec{MD} \quad \text{--- ① addition} \\ &= z + (-iz) \quad \text{--- ①} \\ &= z - iz\end{aligned}$$

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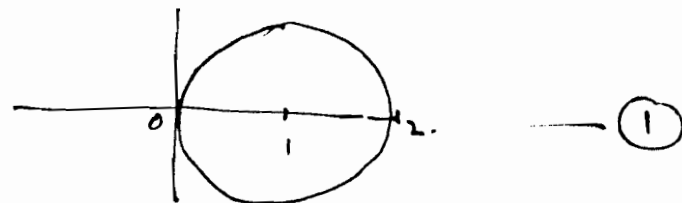


$$\begin{aligned}\text{ii) Volume of slice} &= \pi (R^2 - r^2) \Delta x \\ &= \pi ((y+1)^2 - 1^2) \Delta x \\ &= \pi ((y+1-1)(y+1+1)) \Delta x \\ &= \pi y \cdot (y+2) \Delta x \\ \text{but } y &= (x-1)^2 \\ &= \pi (x-1)^2 ((x-1)^2 + 2) \Delta x \\ &= \pi [(x-1)^4 + 2(x-1)^2] \Delta x\end{aligned}$$

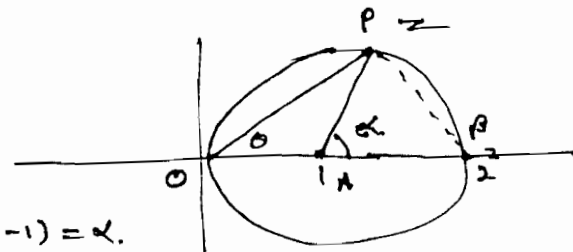
$$\text{Volume} = \lim_{\Delta x \rightarrow 0} \sum_1^2 \pi [(x-1)^4 + 2(x-1)^2] \Delta x$$

$$\begin{aligned}&= \pi \int_1^2 ((x-1)^4 + 2(x-1)^2) dx \quad \text{① idea of annulus} \\ &= \pi \left[\frac{(x-1)^5}{5} + 2 \frac{(x-1)^3}{3} \right]_1^2 \quad \text{① express} \\ &= \pi \left[\frac{1}{5} + \frac{2}{3} - 0 \right] \quad \text{either ①} \\ &= \frac{13\pi}{15}\end{aligned}$$

d) i)



ii)



Let $\arg(z-1) = \alpha$.

$\triangle OPA$ is isosceles — ①

$\therefore \angle OPA = \theta$.

$\therefore \alpha = 2\theta$

$\therefore \arg(z-1) = 2\theta$

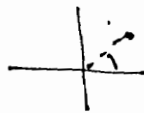
$$\begin{aligned}\text{iii } \text{Arg}(z^2 - 3z + 2) &= \text{Arg}(z-1)(z-2) \\ &= \text{Arg}(z-1) + \text{Arg}(z-2) \quad \text{--- ①} \\ &= 2\theta + \beta \quad (\text{on diagram}) \\ &= 2\theta + \theta + \pi/2 \quad (\text{exterior } \angle \text{ or } \Delta \text{ at angle in semi-circle}) \\ &= 3\theta + \pi/2. \quad \text{--- ①}\end{aligned}$$

version 2

a) i) $z = 2 \operatorname{cis} \pi/6 \rightarrow \textcircled{1} \text{ mod}$
 $\rightarrow \textcircled{1} \text{ arg.}$

version 2

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ii) If z is a solution then

$$(2 \cos \pi/6)^7 + 64(2 \cos \pi/6) \cos$$

$$\text{LHS} = 128 \cos \frac{7\pi}{6} + 128 \cos \frac{\pi}{6}$$

$$= -128 \cos(\pi/6) + 128 \cos \pi/6$$

$\approx R H 1$

b) i) $\frac{1}{2} = \frac{1}{x+iy} \times \frac{x-iy}{x-iy}$

$$= \frac{x - iy}{x^2 + y^2} \quad \text{--- (1)}$$

$$\begin{aligned} \text{ii)} \quad \frac{z + \frac{1}{z}}{2} &= \frac{x + iy + \frac{x - iy}{x^2 + y^2}}{2} \\ &= \frac{(x^2 + y^2)(x + iy) + x + iy}{x^2 + y^2} \end{aligned}$$

$$\operatorname{Re}\left(z - \frac{1}{z}\right) = \frac{x(x^2 + y^2) - x}{x^2 + y^2}$$

$$\therefore \frac{x(x^2+y^2)-x}{x^2+y^2} = 0$$

$$\therefore x(x^2 + y^2) - x = 0$$

$$x(x^2 + y^2 - 1) = 0$$

$$\therefore x=0 \quad \text{or} \quad x^2+y^2=1 \quad x, y \neq 0$$

(2) bath.

with restrictions

$$x, y \neq 0$$

00.5

c). i) $\omega = \text{cis } \pi/4$

$$1. \omega^3 = \cos \frac{3\pi}{4} \quad \text{--- (1)}$$

ii) $1, \omega^2, \omega^4, \omega^6$ are the roots of the equation $z^4 = 1$

$$\therefore 1 + w^2 + w^4 + w^6 = 0 \quad (\text{sum of roots})$$

$$\therefore w^2 + w^4 + w^6 = -1 \quad \text{--- (1) answer (1) reason.}$$

d) i) $\frac{2x}{25} + \frac{2y}{9} \frac{dy}{dx} = 0$ — (I)

$$\begin{aligned}\frac{dy}{dx} &= \frac{-2x}{25} \times \frac{9}{2y} \\ &= \frac{-9x}{25y} \quad \text{--- (1)}\end{aligned}$$

ii) $a(x, y, z) = \frac{dy}{dx} = -\frac{9x}{25y}$

$\therefore$ equat $y - y_1 = \frac{-9x_1}{25y_1} (x - x_1)$

$$\frac{441}{9} - \frac{y_1^2}{9} = -\frac{2x_1}{25} + \frac{x_1^2}{25}$$

$$\therefore \frac{x_1^2}{25} + \frac{y_1^2}{9} = \frac{x_1^2}{25} + \frac{y_1^2}{9} \quad (\text{since } x_1, y_1 \text{ lies on ellipse})$$

$$\therefore \frac{xx_1}{25} + \frac{yy_1}{9} = 1$$

ii) $x = \pm 25/4$ — (1)

iv) when $x = 25/4$

$$\frac{25}{4} \cdot \frac{x_1}{25} + \frac{54}{9} = 1$$

$$\frac{x_1}{4} + \frac{y_1}{9} = 1$$

$$\therefore y = \frac{9(4-x_1)}{4y_1} \quad \text{--- (1)}$$

Now $y < 0$

$$\therefore \frac{9(4-x_1)}{4y_1} < 0$$

but $y_i \geq 0$

$$\therefore 4 - x_1 < x_1 > 4 \quad \text{--- (1)}$$

$$\therefore 4 < x_1 < 5. \quad \text{--- (1)}$$